CS182: Introduction to Machine Learning, Fall 2025
Homework 5

(Due on Monday, Dec. 29 at 11:59 pm (CST))

Notice:

- Please submit your assignments via Gradescope. The entry code is 428YBY.
- Please make sure you select your answer to the corresponding question when submitting your assignments.
- Each person has a total of five days to be late without penalty for all the assignments. Each late delivery less than one day will be counted as one day.

1. [20 points] [*Ensemble Learning*]You are refining the terrain friction prediction module for a Mars rover. To improve generalization and prevent overfitting on noisy sensor data, you decide to add regularization to the standard gradient boosting algorithm.

   At iteration T, the algorithm minimizes a regularized squared error objective function. Let $r_i = y_i - M_{T-1}(x_i)$ be the current residual. We seek the optimal weight β_T for the current weak learner $G_T(x)$ to minimize:

   $$J(\beta_T) = \sum_{i=1}^{n} \frac{1}{2}(r_i - \beta_T G_T(x_i))^2 + \frac{\lambda}{2}\beta_T^2$$

   Here, $\lambda > 0$ is the regularization parameter.

   You are given a mini-batch of 3 training samples. The regularization parameter is set to $\lambda = \mathbf{1.0}$.

Sample ID (i)	**Residual** (r_i)	**Weak Learner Output** $(G_T(x_i))$
1	2.0	1
2	-1.5	-1
3	0.5	1

   (a) Derive the closed-form expression for the optimal coefficient β_T^* that minimizes $J(\beta_T)$. Your expression must include the regularization term λ. Using the data in the table, calculate the exact numerical value of β_T^*. [10 points]

   (b) The robot faces specific environmental constraints. Apply ensemble learning theory to justify the best architecture for each case.[10 points]

   i. Mechanical vibrations introduce random noise into sensor readings, causing individual Decision Trees to exhibit High Variance (instability).
   - Between Bagging and Boosting, which method effectively mitigates this issue?
   - Explain your choice based on the mechanism of "Model Averaging" and Bias-Variance decomposition.

   ii. The robot must switch reliance between a Camera (Day-optimized) and Lidar (Night-optimized).
   - Explain why a Mixture of Experts (MoE) is superior to standard Voting in this scenario.
   - Specifically, describe how the Gating Network calculates the dynamic weights $w_j(x)$ to achieve "Localization" based on environmental light intensity.

Solution:

We differentiate the objective function w.r.t β_T:

$$\frac{\partial J}{\partial \beta_T} = \sum_{i=1}^{n} -(r_i - \beta_T G_T(x_i))G_T(x_i) + \lambda \beta_T$$

Set to 0 to find the minimum:

$$-\sum_{i=1}^{n} r_i G_T(x_i) + \beta_T \sum_{i=1}^{n} G_T(x_i)^2 + \lambda \beta_T = 0$$

$$\beta_T \left(\sum_{i=1}^{n} G_T(x_i)^2 + \lambda \right) = \sum_{i=1}^{n} r_i G_T(x_i)$$

$$\beta_T^* = \frac{\sum_{i=1}^{n} r_i G_T(x_i)}{\sum_{i=1}^{n} G_T(x_i)^2 + \lambda}$$

- **Numerator ($\sum r \cdot G$):**

$$(2.0 \times 1) + (-1.5 \times -1) + (0.5 \times 1)$$
$$= 2.0 + 1.5 + 0.5 = \mathbf{4.0}$$

- **Denominator ($\sum G^2 + \lambda$):**

$$(1^2 + (-1)^2 + 1^2) + 1.0$$
$$= (1 + 1 + 1) + 1.0 = 3.0 + 1.0 = \mathbf{4.0}$$

- **Final Result:**

$$\beta_T^* = \frac{4.0}{4.0} = \mathbf{1.0}$$

(a) High Variance (5 Points):

- **Selection: Bagging** is preferred. (2 pts)
- **Reasoning:** Bagging reduces variance by averaging multiple models trained on bootstrap samples. For unstable learners (high variance) affected by noise, averaging independent errors reduces the ensemble variance by a factor of roughly $1/L$, whereas Boosting might overfit to the noise. (3 pts)

(b) Context-Dependent Sensors (5 Points):

- **MoE vs. Voting:** Standard Voting uses fixed weights for all inputs. MoE is superior because it allows weights to vary based on the input context (Day vs. Night). (2 pts)
- **Gating Network:** The Gating Network takes the input x (light level) and outputs weights $w_j(x)$. It "localizes" the decision by assigning high weight ($w \approx 1$) to the Camera during the day and to the Lidar at night, effectively partitioning the input space among experts. (3 pts)

2. [20 points] [*Ensemble Learning: Gradient Boosting*] Gradient Boosting is a general ensemble learning method that constructs a strong learner by sequentially adding weak learners. Unlike ensemble methods that train weak learners independently, Gradient Boosting builds the model iteratively: at each iteration, a new weak learner is added to improve the current model's performance.

   The key idea of Gradient Boosting is to formulate learning as an optimization problem. Given a loss function, the algorithm seeks a prediction function that minimizes the empirical risk on the training data. This is achieved by performing gradient descent in function space: at each iteration, a weak learner is trained to approximate the negative gradient of the loss function with respect to the current model's predictions, and is then added to the ensemble with an appropriate coefficient.

   Formally, at iteration t, the metalearner has the additive form

$$M_t(x) = \sum_{k=1}^{t} \beta_k h_k(x),$$

   where $h_k(\cdot)$ is the k-th weak learner and $\beta_k \in \mathbb{R}$ is its corresponding coefficient. We consider a binary classification problem with training data

$$(x_i, y_i)_{i=1}^n, \quad x_i \in \mathbb{R}^d, \quad y_i \in \{+1, -1\},$$

   which minimizes the logistic loss:

$$L(y, \rho) = \log\big(1 + \exp(-y\rho)\big),$$

   where $\rho = M_t(x)$.

   (a) Write down the empirical risk minimized by the Gradient Boosting metalearner after T iterations. Your answer should be expressed explicitly in terms of $\beta_1, \ldots, \beta_T$, $h_1, \ldots, h_T$, and the training data. [7 points]

   (b) Assume that at iteration T, the weak learner h_T has already been trained and all previous coefficients $\beta_1, \ldots, \beta_{T-1}$ are fixed constants. Derive the expression for the gradient of the empirical risk with respect to β_T . Your answer should be written in terms of $M_{T-1}(x_i), h_T(x_i)$, and y_i. [13 points]

Solution:

   (a) After T iterations, the metalearner is

$$M_T(x) = \sum_{k=1}^{T} \beta_k h_k(x).$$

   The logistic loss is defined as

$$L(y, \rho) = \log\big(1 + \exp(-y\rho)\big), \quad \rho = M_T(x).$$

   Therefore, the total empirical risk is

$$\mathcal{R}_T = \frac{1}{n}\sum_{i=1}^{n} \log\Big(1 + \exp\big(-y_i M_T(x_i)\big)\Big) = \frac{1}{n}\sum_{i=1}^{n} \log\Big(1 + \exp\big(-y_i \sum_{k=1}^{T} \beta_k h_k(x_i)\big)\Big)$$

   (b) At iteration T , write the model output as

$$M_T(x_i) = M_{T-1}(x_i) + \beta_T h_T(x_i),$$

   where M_{T-1} is fixed.
   The empirical risk becomes

$$\mathcal{R}_T = \frac{1}{n}\sum_{i=1}^{n} \log\Big(1 + \exp\big(-y_i(M_{T-1}(x_i) + \beta_T h_T(x_i))\big)\Big).$$

   Derivative for a Single Sample. We define

$$\ell_i(\beta_T) = \log\Big(1 + \exp\big(-y_i(M_{T-1}(x_i) + \beta_T h_T(x_i))\big)\Big).$$

Taking derivative w.r.t. β_T :

$$\frac{\partial \ell_i}{\partial \beta_T} = \frac{\exp(-y_i \rho_i)}{1 + \exp(-y_i \rho_i)} \cdot (-y_i h_T(x_i)),$$

where

$$\rho_i = M_{T-1}(x_i) + \beta_T h_T(x_i).$$

Equivalently,

$$\frac{\partial \ell_i}{\partial \beta_T} = -y_i h_T(x_i) \cdot \sigma(-y_i \rho_i),$$

with $\sigma(z) = \frac{1}{1+\exp(-z)}$ being the sigmoid function.
Gradient of the Empirical Risk

$$\frac{\partial \mathcal{R}_T}{\partial \beta_T} = -\frac{1}{n} \sum_{i=1}^{n} y_i h_T(x_i) \cdot \frac{1}{1 + \exp\left(y_i (M_{T-1}(x_i) + \beta_T h_T(x_i))\right)}$$

3. [20 points] [*Hypothesis Testing*] Consider the (binary) hypothesis testing, where there are only two hypotheses to discriminate between given our observation y:

$$H_0 : Y \sim P_{Y|X=0} \text{ (the null hypothesis)}$$
$$H_1 : Y \sim P_{Y|X=1} \text{ (the alternate hypothesis)}$$

Define a decision rule/test $\phi : \mathcal{Y} \mapsto \{0, 1\}$. Recall that

- Type I Error (False Positive) Probability: $\Pr[\phi(Y) = 1|X = 0]$
- Type II Error (False Negative) Probability: $\Pr[\phi(Y) = 0|X = 1]$

The goal to choose a test that minimizes the type II error probability subject to a constraint on the type I error probability. That is, for some $\alpha \geq 0$, find

$$\phi^*_\alpha = \underset{\phi:\Pr[\phi(Y)=1|X=0]\leq\alpha}{\arg\min} \Pr[\phi(Y) = 0|X = 1] = \underset{\phi:\Pr[\phi(Y)=1|X=0]\leq\alpha}{\arg\max} \Pr[\phi(Y) = 1|X = 1] \tag{1}$$

Definition 1 ((randomized) threshold test)**.** *A **(randomized) threshold test** is a test of the following form:*

$$\phi(y) = \begin{cases} 0 & L(y) < \lambda \\ Bernoulli(\gamma) & L(y) = \lambda \\ 1 & L(y) > \lambda \end{cases}$$

where $L(y) = \frac{P_{Y|X}(y|1)}{P_{Y|X}(y|0)}$ *is the **likelihood ratio** and* λ, γ *are some constants.*

(a) Show that there exists a (randomized) threshold test ϕ such that

$$\Pr[\phi(Y) = 1|X = 0] = \mathrm{E}[\phi(Y)|X = 0] = \alpha \tag{2}$$

for any $\alpha \in [0, 1]$.[5 points]

(b) Show that if a test is a threshold test with (2) holds, then, it is an optimal solution of (1) [10 points]

(c) Consider the Gaussian example with

$$P_{Y|X=0} \sim \mathcal{N}(0, \sigma^2)$$
$$P_{Y|X=1} \sim \mathcal{N}(1, \sigma^2)$$

Let Φ denote the CDF of the standard normal distribution. Find an optimal solution of (1) and the relationship between γ, λ and α. [5 points]

Solution:

(a) *Proof.* For $\alpha = 0$ and $\alpha = 1$ the theorem is easily seen to be true provided the value $\lambda = +\infty$ and $0 \cdot \infty$ is interpreted as 0. Throughout the proof we shall therefore assume $0 < \alpha < 1$.
Let $G(\lambda) = \Pr[L(Y) > \lambda|X = 0]$. It is trivial that $1 - G(\lambda)$ is a cumulative distribution function, and hence $G(\lambda)$ is non-increasing and continuous on the right,$G(-\infty) = 1$, and $G(\infty) = 0$
Let $\hat{\lambda} = \inf\{\lambda \in \mathbb{R} : G(\lambda) \leq \alpha\}$

- If $G(\hat{\lambda}) = \alpha$, set $\gamma = 0$
- If $G(\hat{\lambda}) < \alpha$, set $\gamma = \frac{\alpha - G(\hat{\lambda})}{G(\hat{\lambda}_-) - G(\hat{\lambda})} = \frac{\alpha - G(\hat{\lambda})}{\Pr[L(Y)=\hat{\lambda}|X=0]}$

Both cases ensure that $\Pr[\phi(Y) = 1|X = 0] = \alpha$ □

(b) *Proof.* Suppose that ϕ is a threshold test with (2) holds.
Let ϕ^* be any other test with $\Pr[\phi^*(Y) = 1|X = 0] \leq \alpha$.
Define the sets

$$S^+ = \{y : \phi(y) - \phi^*(y) > 0\},$$
$$S^- = \{y : \phi(y) - \phi^*(y) < 0\}.$$

If $y \in S^+$, then $\phi(y) > \phi^*(y)$ and $\phi(y) = 1$, which implies $L(y) \geq \lambda$.
If $y \in S^-$, then $\phi(y) < \phi^*(y)$ and $\phi(y) = 0$, similarly, we have $L(y) \leq \lambda$.
Hence, on $S = S^+ \cup S^-$,

$$\big(\phi(y) - \phi^*(y)\big)\big(L(y) - \lambda\big) \geq 0.$$

$$\big(\phi(y) - \phi^*(y)\big)\big(p_1(y) - \lambda p_0(y)\big) \geq 0.$$

where $p_0(y) = P_{Y|X}(y|0), p_1(y) = P_{Y|X}(y|1)$
Consequently,

$$\int (\phi - \phi^*)(p_1 - \lambda p_0)\, dy = \int_{S^+ \cup S^-} (\phi - \phi^*)(p_1 - \lambda p_0)\, dy \geq 0.$$

Therefore,

$$\begin{aligned}
\Pr[\phi(Y) = 1|X = 1] - \Pr[\phi^*(Y) = 1|X = 1] &= \int (\phi - \phi^*) p_1 dy \\
&\geq \lambda \int (\phi - \phi^*) p_0 dy \\
&= \lambda(\alpha - \Pr[\phi^*(Y) = 1|X = 0]) \geq 0
\end{aligned}$$

which completes the proof. □

(c) We first compute

$$L(y) = \frac{P_{Y|X}(y|1)}{P_{Y|X}(y|0)} = \exp(\frac{y}{\sigma^2} - \frac{1}{2\sigma^2})$$

Note that $\Pr[L(Y) = \lambda|X = 0] = 0$ for any λ because Y, and hence $L(y)$, is a continuous random variable. So, no randomization to worry about in this case and we can effectively ignore γ.
For λ, set

$$\begin{aligned}
\alpha &= \Pr[\phi(Y) = 1|X = 0] \\
&= \Pr[L(Y) \geq \lambda|X = 0] \\
&= \Pr[\exp(\frac{Y}{\sigma^2} - \frac{1}{2\sigma^2}) \geq \lambda|X = 0] \\
&= \Pr[\frac{Y}{\sigma} \geq \frac{1}{2\sigma} + \sigma \log \lambda|X = 0] \\
&= 1 - \Phi(\frac{1}{2\sigma} + \sigma \log \lambda)
\end{aligned}$$

From (b), we know that a threshold test with the above equation holds forms an optimal solution.

4. [20 points] [*Hypothesis Testing*] In a binary classification task (e.g., click-through rate prediction or fraud detection), two classifiers, Model A and Model B, are evaluated using 5-fold cross-validation. The validation error rates obtained on each fold are reported below:

Fold i	Model A error $p_i^{(A)}$	Model B error $p_i^{(B)}$
1	0.18	0.15
2	0.20	0.17
3	0.16	0.18
4	0.19	0.16
5	0.17	0.14

Assume that the validation sets are sufficiently large so that the error rates can be approximated as normally distributed.

(a) Compute the sample mean and sample standard deviation of the 5-fold cross-validation error rates for Model A. Then construct a 95% two-sided confidence interval for its true mean error rate. [6 points]

(b) Define the per-fold difference

$$d_i = p_i^{(A)} - p_i^{(B)}, \quad i = 1, \ldots, 5.$$

Using a paired t-test, test at significance level $\alpha = 0.05$ whether the two models have the same expected error rate. Clearly state the null hypothesis, test statistic, and conclusion. [8 points]

(c) In practice, suppose you only care whether Model B is better than Model A (i.e., has a lower error rate).

- 1. Formulate an appropriate one-sided hypothesis test.
- 2. Explain why a one-sided test is more suitable than a two-sided test in this scenario.
- 3. Compared with a two-sided test at the same significance level, does the rejection region become larger or smaller?

[6 points]

Solution:

(a) The sample mean of Model A is

$$\bar{p}_A = \frac{1}{5}(0.18 + 0.20 + 0.16 + 0.19 + 0.17) = 0.18.$$

The sample variance is

$$S_A^2 = \frac{1}{4}\sum_{i=1}^{5}(p_i^{(A)} - \bar{p}_A)^2 = \frac{0.0010}{4} = 0.00025,$$

so the sample standard deviation is $S_A = \sqrt{0.00025} \approx 0.0158$. Using the t-distribution with 4 degrees of freedom, $t_{0.025,4} \approx 2.776$. Thus the 95% confidence interval is

$$\bar{p}_A \pm t_{0.025,4}\frac{S_A}{\sqrt{5}} = 0.18 \pm 0.0196 = (0.160,\ 0.200).$$

(b) The per-fold differences are

$$d = [0.03,\ 0.03,\ -0.02,\ 0.03,\ 0.03].$$

Their sample mean is

$$\bar{d} = \frac{0.10}{5} = 0.02,$$

and the sample standard deviation is

$$S_d = \sqrt{\frac{1}{4}\sum_{i=1}^{5}(d_i - \bar{d})^2} = \sqrt{0.0005} \approx 0.0224.$$

We test

$$H_0 : \mu_d = 0 \quad \text{vs.} \quad H_1 : \mu_d \neq 0.$$

The paired t-statistic is

$$t = \frac{\sqrt{5}\,\bar{d}}{S_d} = \frac{2.236 \times 0.02}{0.0224} = 2.00.$$

Since $|t| < t_{0.025,4} = 2.776$, we fail to reject H_0. Therefore, the difference between the two models is not statistically significant at the 5% level.

(c)

1. Since $d_i = p_i^{(A)} - p_i^{(B)}$, Model B is better when $d_i > 0$. Thus, the one-sided hypotheses are

$$H_0 : \mu_d \leq 0, \quad H_1 : \mu_d > 0.$$

– 2. In practice, we only care about improvement. A one-sided test directly reflects this directional objective and has higher power to detect performance gains.

– 3. At the same significance level, a one-sided test allocates all probability mass to one tail, resulting in a smaller critical value and a larger rejection region in the direction of interest.

5. [20 points] [*Interval Estimation*] Let $X_1, X_2, \ldots, X_n$ be a random sample from the uniform distribution $R(0, \theta)$, where $\theta > 0$ is an unknown parameter. Denote $M_n = \max\{X_1, X_2, \ldots, X_n\}$.

   (a) [8 points] Find the probability density function $f_{M_n}(t)$ and the cumulative distribution function $F_{M_n}(t)$ of M_n.

   (b) [6 points] For a given confidence level $1-\alpha$ $(0 < \alpha < 1)$, find a constant $c_n > 1$ such that the interval $[M_n, c_n M_n]$ is a confidence interval for θ with confidence coefficient $1-\alpha$. Express c_n explicitly in terms of α and n.

   (c) [6 points] Suppose a sample of size $n = 10$ is drawn from $R(0, \theta)$, yielding the following observed values:

   $$0.2,\ 0.5,\ 0.8,\ 1.2,\ 0.3,\ 0.6,\ 0.9,\ 1.1,\ 0.4,\ 0.7.$$

   Taking the confidence coefficient $1-\alpha = 0.95$ and using the formula derived in part (b), compute the numerical lower and upper bounds of this confidence interval (round your results to three decimal places).

Solution:

(a) Since $X_i \overset{\text{i.i.d.}}{\sim} R(0, \theta)$, the common pdf and cdf are

$$f_X(x) = \frac{1}{\theta}\mathbb{I}_{(0,\theta)}(x), \quad F_X(x) = \frac{x}{\theta} \text{ for } 0 < x < \theta.$$

Since $X_1, X_2, \ldots, X_n$ are independent and identically distributed, the event $\{M_n \le t\}$ is equivalent to $\{X_1 \le t, X_2 \le t, \ldots, X_n \le t\}$. By independence,

$$F_{M_n}(t) = \mathbb{P}(M_n \le t) = \mathbb{P}(X_1 \le t) \cdot \mathbb{P}(X_2 \le t) \cdots \mathbb{P}(X_n \le t) = [F_X(t)]^n.$$

The cdf of M_n is

$$F_{M_n}(t) = [F_X(t)]^n = \begin{cases} 0, & t \le 0, \\ \left(\dfrac{t}{\theta}\right)^n, & 0 < t < \theta, \\ 1, & t \ge \theta. \end{cases}$$

Differentiating for $0 < t < \theta$ gives the pdf:

$$f_{M_n}(t) = \frac{n t^{n-1}}{\theta^n}, \quad 0 < t < \theta,$$

and $f_{M_n}(t) = 0$ otherwise.

(b) We require $\mathbb{P}(M_n \le \theta \le c_n M_n) = 1 - \alpha$. Since $\theta \ge M_n$ always holds, this probability equals $\mathbb{P}(\theta \le c_n M_n)$. Equivalently,

$$\mathbb{P}\left(M_n \ge \frac{\theta}{c_n}\right) = 1 - \alpha.$$

Using the cdf from part (a),

$$\mathbb{P}\left(M_n \ge \frac{\theta}{c_n}\right) = 1 - F_{M_n}\left(\frac{\theta}{c_n}\right) = 1 - \left(\frac{\theta/c_n}{\theta}\right)^n = 1 - c_n^{-n}.$$

Setting $1 - c_n^{-n} = 1 - \alpha$ yields

$$c_n^{-n} = \alpha \quad \Rightarrow \quad c_n = \alpha^{-1/n}.$$

Hence, the desired confidence interval is $\left[\, M_n,\ \alpha^{-1/n} M_n \,\right]$.

(c) From the data, $M_n = 1.2$. With $1 - \alpha = 0.95$, we have $\alpha = 0.05$ and $n = 10$. Thus,

$$c_n = (0.05)^{-1/10} = (0.05)^{-0.1}.$$

Compute:

$$\ln c_n = -0.1 \ln(0.05) \approx -0.1 \times (-2.9957) = 0.29957, \quad \text{so } c_n \approx e^{0.29957} \approx 1.349.$$

The confidence interval bounds are:

$$\text{Lower bound} = M_n = 1.200, \qquad \text{Upper bound} = c_n M_n \approx 1.2 \times 1.349 \approx 1.619.$$

Therefore, the approximate 95% confidence interval is

$$[\,1.200,\ 1.619\,].$$